

Module 1

SAP-1 Architecture

The SAP-1 design contains the basic necessities for a functional microprocessor. In this laboratory activity, a Simple-As-Possible (SAP)-1 computer will be designed in Verilog HDL.

Learning objectives:

- To design an 8-bit computer that can perform simple arithmetic operations using the SAP-1 architecture
- To design the processing unit, control unit, memory to store data and instructions, and a central bus
- To perform a simple algorithm such as instruction execution cycle to process the instruction of a program

1. Registers (Part 2)

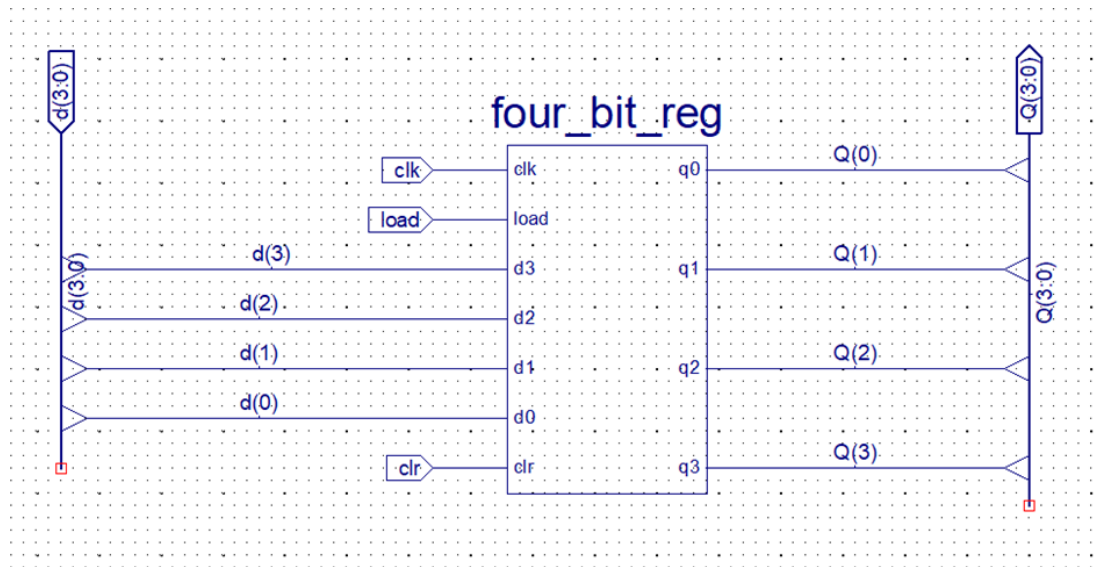
To be able to store a byte (8-bits) of data, a combination of eight flip-flops is needed to work as a unit. A register is a circuit with two or more D flip-flops connected together to work exactly the same way it holds data and can be synchronized by the same clock and enable signals. Each flip-flops that is grouped together to form a register can hold different bit and is synchronized by the same clock cycle.

In this activity an 8-bit serial-to-parallel load shift register and serial-to-parallel and parallel-to-serial shift registers will be implemented as a compact single chip design.

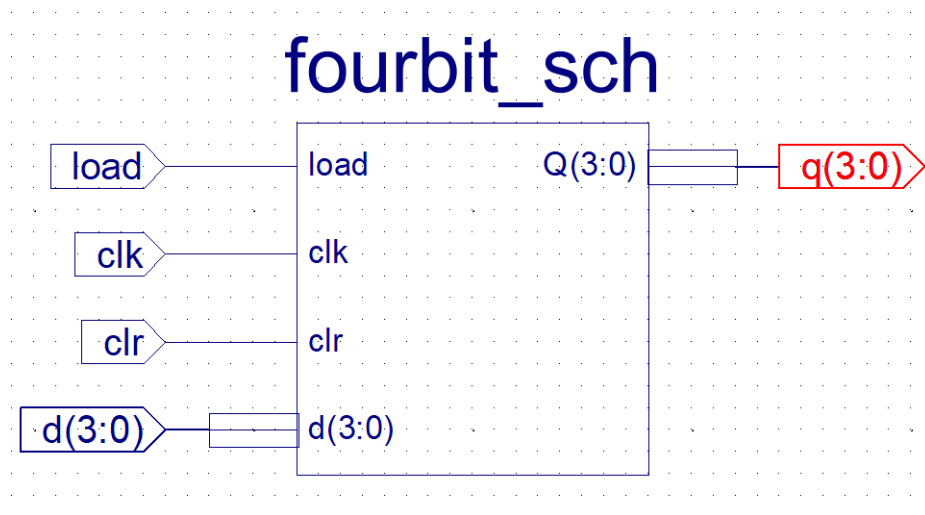
Exercises

Exercises 1: Design implementation

1. Under **Design Utilities**, double click on **Check Design Rules** for the newly created circuit schematic
2. Also, click on **Synthesize – XST** to check for errors



3. If no errors found in the circuit schematic, **Create Schematic Symbol** under *Design Utilities*
4. Create new schematic for the **Final IC design** by adding **New Source -> Schematic**



Final IC design for
4-bit Register with parallel load and asynchronous clear

5. Simulate the behavior of the newly created IC. (New Source -> Verilog Test-Fixtures)

Design Constraints

1. Design the following Registers
 - a. 8-bit Serial-to-Parallel Shift Register

0	No Change
1	One bit from Serial_in is shifted in

- b. 8-bit Serial-to-Parallel and Parallel-to-Serial Shift Register

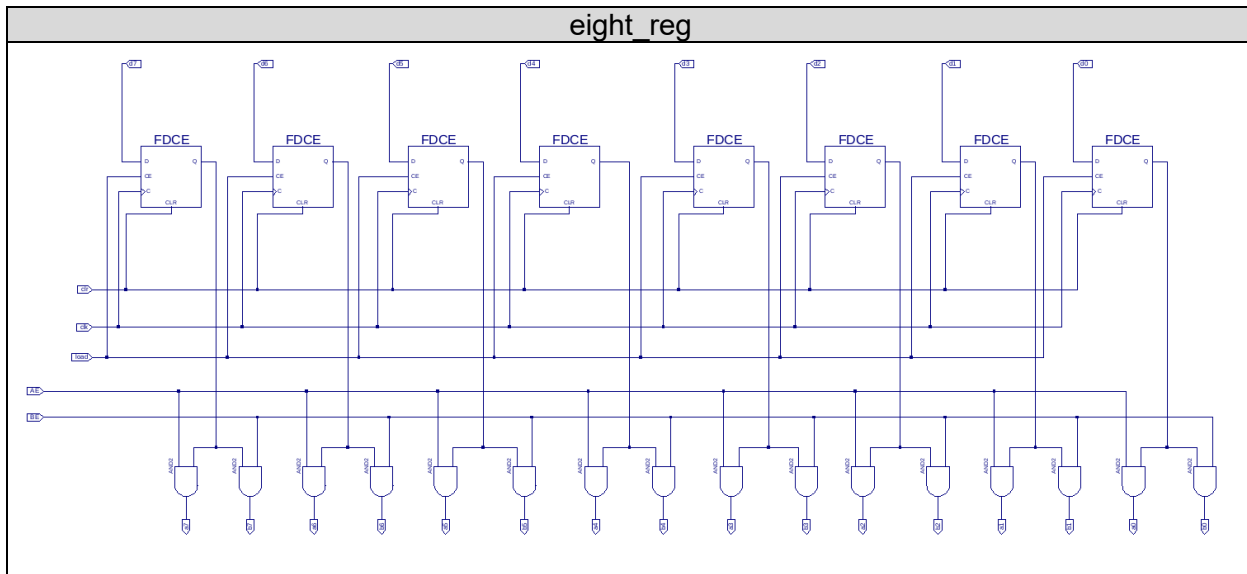
SHSel1	SHSel0	Operation
0	0	No Operation
0	1	Parallel Load in new value
1	0	Shift Right
1	1	Rotate Right

Note: Refer to Chapter 8: Enoch Hwang

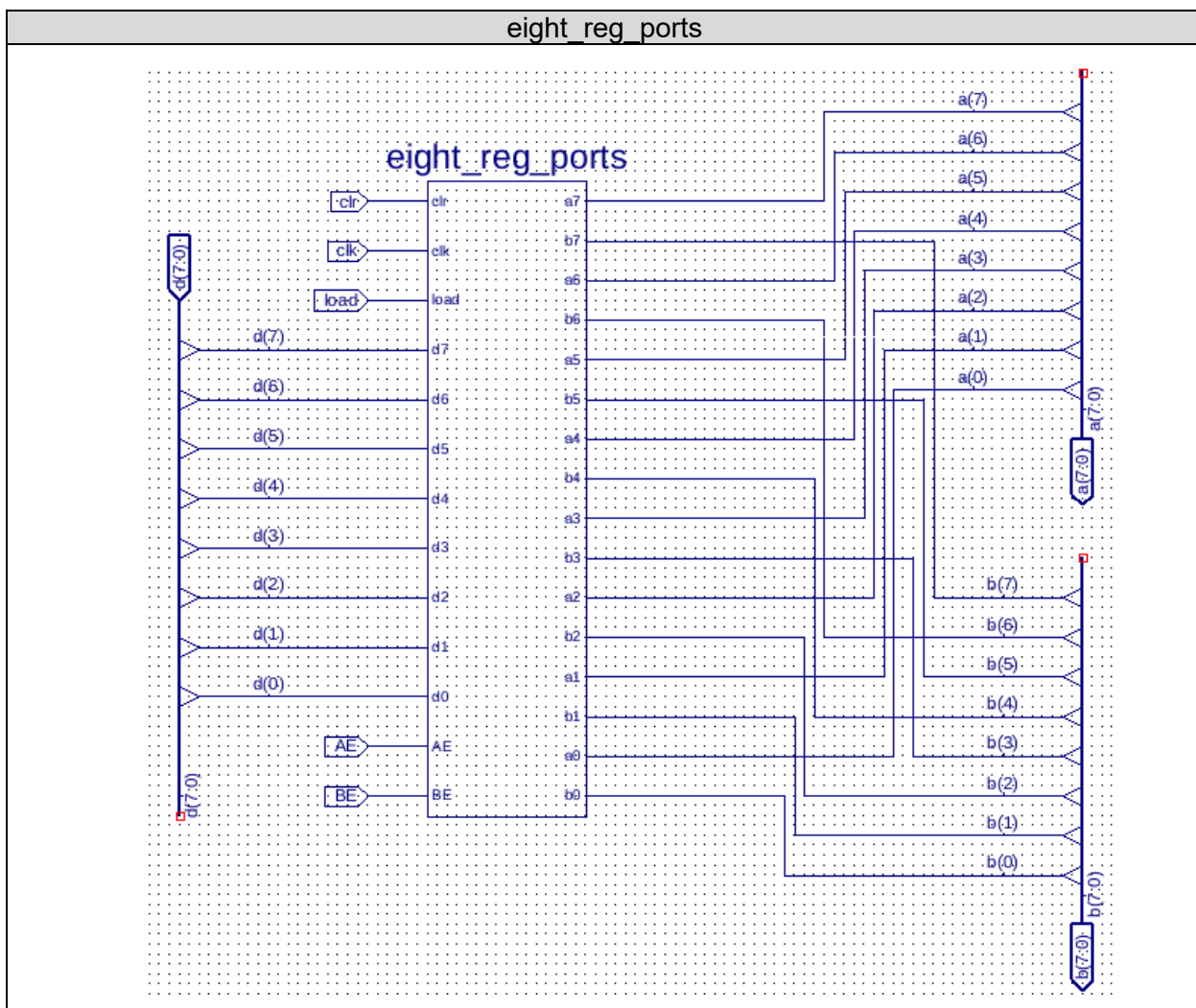
Constraints:

This image shows a single sheet of white paper with horizontal blue or grey ruling lines. The lines are evenly spaced and run across the width of the page. There are no margins, text, or other markings on the paper.

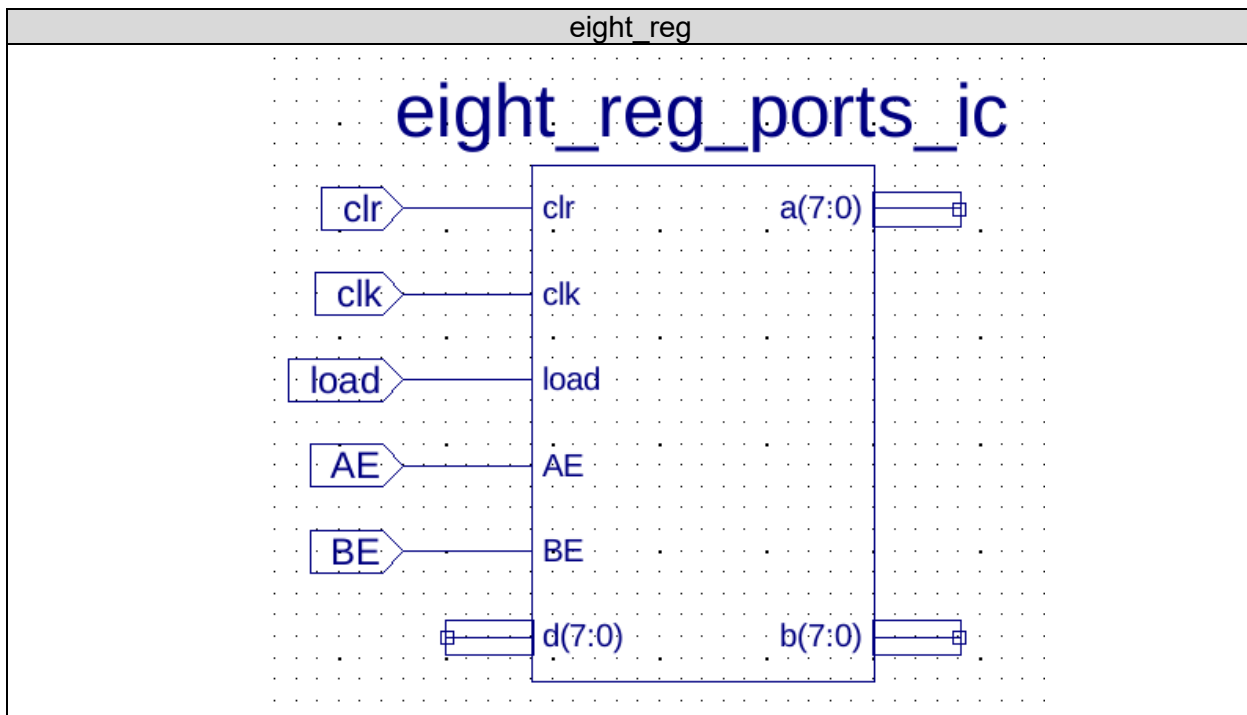
1. Eight-Bit Register



2. Eight-Bit Register Ports IC



3. Eight-Bit Register Ports IC (Vectorized)



4. Test Bench

```
eight_reg
// Verilog test fixture created from schematic
/home/ise/four_eight_reg/eight_reg_ports_ic.sch - Sun Jan 18 13:25:19
2026

`timescale 1ns / 1ps

module eight_reg_ports_ic_eight_reg_ports_ic_sch_tb();

// Inputs
reg clr;
reg clk;
reg load;
reg AE;
reg BE;
reg [7:0] d;

// Output
wire [7:0] a;
wire [7:0] b;

// Bidirs

// Initialize Inputs
initial
begin
    clr = 1'b1;
    load = 1'b0;
```

```

        AE = 1'b0; BE = 1'b0;
    end

    initial
    begin
        #5 clr = 1'b0;
        load = 1'b1;
        AE = 1'b1; BE = 1'b0;

        #10 d = 8'b0010_1100;
    end

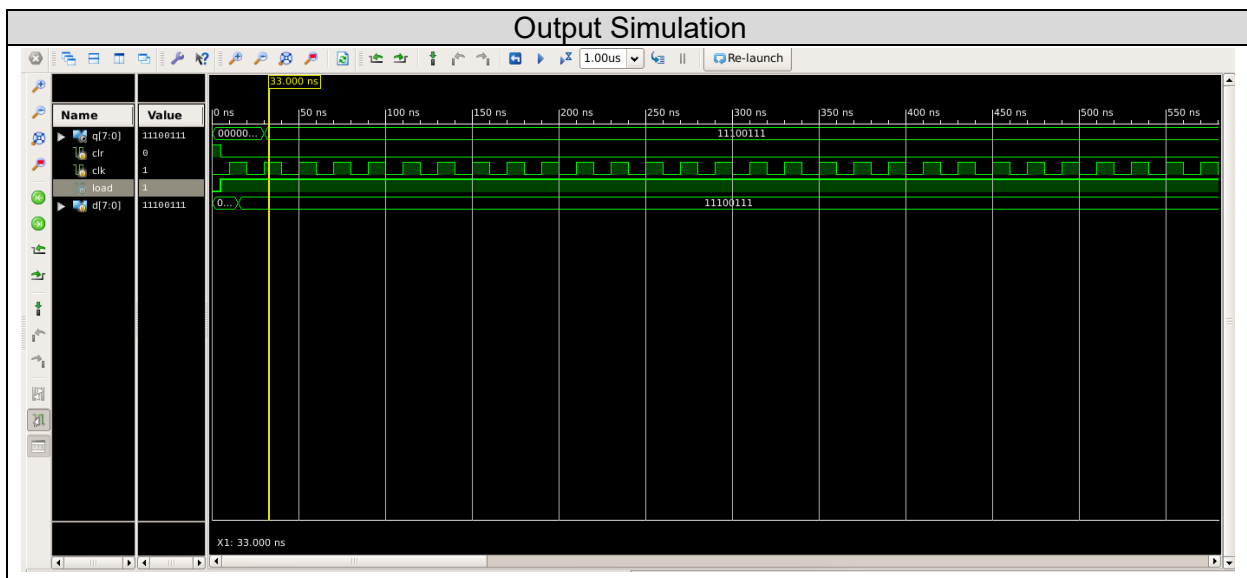
    initial
    begin
        clk = 1'b0;
        forever
        #10 clk = ~clk;
    end

    // Instantiate the UUT
    eight_reg_ports_ic UUT (
        .clr(clr),
        .clk(clk),
        .load(load),
        .AE(AE),
        .BE(BE),
        .d(d),
        .a(a),
        .b(b)
    );

endmodule

```

5. Output Simulation



References:

Digital Computer Electronics, 3rd Edition; Malvino, Albert Paul; Brown, Jerald A., McGraw-Hill, 1993

PCB Design Example:

<https://www.youtube.com/watch?v=iEPjIELWarw&list=PLbUnUWOWn7M8QmODC1wdVhVPIPu4FwBc&index=1>

EasyEDA Dowload page

<https://easyeda.com/page/download>

SAP-1 Design

<https://www.youtube.com/watch?v=oJvJXJb7uEU&list=PLbUnUWOWn7M8QmODC1wdVhVPIPu4FwBc&index=2>

<https://www.youtube.com/watch?v=VzVTrh5uBjQ&list=PLbUnUWOWn7M8QmODC1wdVhVPIPu4FwBc&index=3>

<https://www.youtube.com/watch?v=6xDo0wK8xAY&list=PLbUnUWOWn7M8QmODC1wdVhVPIPu4FwBc&index=4>

Build an 8-bit computer from scratch

<https://eater.net/8bit>

Simulator

<https://circuitverse.org/simulator>

SAP-1 Processor Architecture

<https://dangrie158.github.io/SAP-1/index.html>

<https://deeprajbhujel.blogspot.com/2015/12/sap-2-instruction-sets.html>

Digital Logic and Microprocessor Design with VHDL by Enoch O. Hwang